

Jonathan Huang

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EDUCATION

Northwestern University Feinberg School of Medicine <i>M.D. (Medical Scientist Training Program)</i>	2020 - 2027 <i>Chicago, IL</i>
Northwestern University <i>Ph.D., Biomedical Engineering</i> Advised by Mozziyar Etemadi, MD PhD; GPA: 4.0/4.0	2020 - 2025 <i>Chicago, IL</i>
Brown University <i>Sc.B., Neuroscience and Computer Science, magna cum laude</i> Awarded with Honors in Neuroscience; GPA: 4.0/4.0	2016 - 2020 <i>Providence, RI</i>

AWARDS & HONORS

• Forbes 30 Under 30 (Healthcare)	2026
• “Distinguished” class category, Feinberg School of Medicine (top quartile of class)	2026
• Trainee Research Prize, Radiological Society of North America	2025
• Research Intensive Scholarly Emphasis (RISE) Scholar, Feinberg School of Medicine	2022
• John P. Donoghue Prize in Neuroscience, Brown University	2020
• Elected to Sigma Xi	2020
• Karen T. Romer Undergraduate Teaching and Research Award, Brown University	2019
• National Merit Scholarship	2016

PROFESSIONAL EXPERIENCE

Co-Founder & Chief Scientific Officer <i>MedikaMind</i> Co-founded and lead scientific efforts for a startup company spun out of Northwestern Medicine to continue development and distribution of generative AI radiology reporting tools	2026 - <i>Chicago, IL</i>
Medical Scribe <i>Brown Emergency Medicine</i> Documented patient encounters for Emergency Department providers	2019 - 2020 <i>Providence, RI</i>

RESEARCH EXPERIENCE

Research Assistant <i>Healthcare Integrated Technologies Lab, Northwestern Medicine</i> Development, implementation, and prospective evaluation of clinical machine learning tools relating to vision/language tasks in radiology and ophthalmology under the supervision of Dr. Mozziyar Etemadi	2020 - 2025
Research Assistant <i>Barnea Lab, Brown University</i> Honors thesis: “Genetic Targeting of Precise Intersectional Neuronal Populations Using <i>trans</i>-Tango” under the supervision of Dr. Gilad Barnea	2018 - 2020

TEACHING

Teaching Assistant BME 305: Quantitative Systems Physiology <i>Department of Biomedical Engineering, Northwestern University</i>	Winter 2025
Teaching Assistant NEUR 1030: Neural Systems <i>Department of Neuroscience, Brown University</i>	Fall 2018 & 2019
Teaching Assistant CSCI 0220: Introduction to Discrete Structures and Probability <i>Department of Computer Science, Brown University</i>	Spring 2019

LEADERSHIP

Clinic Chief Education-Centered Medical Home • Served as student chief of a clinic serving primary care and transitional care needs.	2021-2022
President & Founder Writes of Passage • Founded and led a student group focused on medical journalism and outreach for medical professionals	2020-2022
President Neurological Surgery Interest Group • Led a group of medical students interested in neurosurgery by planning events and learning opportunities	2021-2022

SERVICE

Volunteer CommunityHealth Clinic Chicago • Roomed, screened, and evaluated patients in a clinic providing comprehensive eye care for uninsured patients	2026 -
Volunteer Chinatown Free Clinic • Volunteered at community health fairs and screening events serving underinsured patients, working alongside on-site physicians	2020 - 2023
Mentor NU-Feinberg Connections • Provided longitudinal guidance to three pre-medical undergraduate students on coursework, research, and the medical school application process	2020 - 2022

GRANT FUNDING

Investigator “Physician Attitudes Towards Artificial Intelligence and Result Management” <i>Center for Bioethics and Medical Humanities & Institute for Augmented Intelligence in Medicine</i>	2022
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INTELLECTUAL PROPERTY

1. “Assessment of clinical evaluations from machine learning systems”
U.S. Patent [12,505,917](#), Dec. 23, 2025
2. “Assessment of clinical evaluations from machine learning systems”
U.S. Patent [12,450,887](#), Oct. 21, 2025

Articles Under Review:

1. **Huang J**, Bucciarelli NJ, Tanna AP. Increasing Prevalence of Unverifiable References in the Ophthalmology Literature in the Large Language Model Era. Under review in *Ophthalmology*.

Peer-Reviewed Original Research:

1. **Huang J**, Wittbrodt MT, Teague CN, et al. Efficiency and Quality of Generative AI-Assisted Radiograph Reporting. *JAMA Network Open*. 2025;8(6):e2513921. [doi:10.1001/jamanetworkopen.2025.13921](https://doi.org/10.1001/jamanetworkopen.2025.13921)
2. Rabin EE, Reyes SG, Chandrasekar S, Huang B, **Huang J**, et al. Minimally Invasive Transforaminal Lumbar Interbody Fusion: Outcomes and Complications of Single and Multilevel Fusions in Younger and Older Adults. *Spine Open*. 2025;1(2):e0037. [doi:10.1097/bn9.0000000000000037](https://doi.org/10.1097/bn9.0000000000000037)
3. Kaklamanos E, Kristinsdottir K, **Huang J**, et al. From Scope to Script: An Automated Report Generation Model for Gastrointestinal Endoscopy. *MICCAI Workshop on Deep Generative Models*. 2025;16128:131-140. [doi:10.1007/978-3-032-05472-2_13](https://doi.org/10.1007/978-3-032-05472-2_13)
4. **Huang J**, Galal G, Mukhin V, Etemadi M, Tanna A. Prediction and detection of glaucomatous visual field progression using deep learning on macular OCT. *Journal of Glaucoma*. 2024;33(4):246-253. [doi:10.1097/IJG.0000000000002359](https://doi.org/10.1097/IJG.0000000000002359)
5. Kim E, Stec M, Shaikh N, **Huang J**, Ranaivo HR, Mets-Halgrimson R. Refractive changes in children in the Chicago area during the COVID-19 pandemic: a retrospective observational study. *British Journal of Ophthalmology*. 2025;109:543-548. [doi:10.1136/bjo-2024-325984](https://doi.org/10.1136/bjo-2024-325984)
6. Rabin EE, **Huang J**, Kim M, et al. Age-stratified comorbid and pharmacologic analysis of patients with glioblastoma. *Brain, Behavior, & Immunity - Health*. 2024;38:100753. [doi:10.1016/j.bbih.2024.100753](https://doi.org/10.1016/j.bbih.2024.100753)
7. Ellis EM, Drumm MR, Rai S, **Huang J**, et al. Patterns of Antiseizure Medication Use Following Meningioma Resection: A Single-Institution Experience. *World Neurosurgery*. 2024;181:e392-e398. [doi:10.1016/j.wneu.2023.10.068](https://doi.org/10.1016/j.wneu.2023.10.068)
8. **Huang J**, Neill L, Wittbrodt M, et al. Generative Artificial Intelligence for Chest Radiograph Interpretation in the Emergency Department. *JAMA Network Open*. 2023;6(10):e2336100. [doi:10.1001/jamanetworkopen.2023.36100](https://doi.org/10.1001/jamanetworkopen.2023.36100)
9. **Huang J**, Rossen J, Rahmani B, Mets-Halgrimson R. Pediatric Eyelid and Canalicular Lacerations: Epidemiology and Outcomes. *Journal of Pediatric Ophthalmology and Strabismus*. 2023;60(1):33-38. [doi:10.3928/01913913-20220321-02](https://doi.org/10.3928/01913913-20220321-02)
10. Dombrovsky D, Gajjar AA, **Huang J**, Barpujari A, Singh R, Patel NP. Cross-sectional analysis of neurosurgical residency websites during the virtual interview cycle. *World Neurosurgery*. 2023;180:e158-e162. [doi:10.1016/j.wneu.2023.09.018](https://doi.org/10.1016/j.wneu.2023.09.018)
11. Ellis EM, Drumm MR, Rai S, **Huang J**, et al. Long-term antiseizure medication use in patients after meningioma resection: identifying predictors for successful weaning and failures. *Journal of Neuro-Oncology*. 2023;165(1):201-207. [doi:10.1007/s11060-023-04481-6](https://doi.org/10.1007/s11060-023-04481-6)

12. **Huang J**, Basith S, Patel S, Weisman A, Brickman W, Mets M, Mets-Halgrimson R. Ocular Findings in Pediatric Turner Syndrome. *Ophthalmic Genetics*. 2022;43(4). [doi:10.1080/13816810.2022.2045512](https://doi.org/10.1080/13816810.2022.2045512)
13. Domingo J, Galal G, **Huang J**, Soni P, Mukhin V, et al. Preventing Delayed and Missed Care by Applying Artificial Intelligence to Trigger Relevant Imaging Follow-Up. *NEJM Catalyst Innovations in Care Delivery*. 2022;3(4). [doi:10.1056/CAT.21.0469](https://doi.org/10.1056/CAT.21.0469)

Review Articles:

1. Alleman K, Knecht E, **Huang J**, Zhang L, Lam S, DeCuyper M. Multimodal Deep Learning-Based Prognostication in Glioma Patients: A Systematic Review. *Cancers*. 2023;15(2):545. [doi:10.3390/cancers15020545](https://doi.org/10.3390/cancers15020545)
2. Shlobin NA, **Huang J**, Wu C. Learning curves in robotic neurosurgery: a systematic review. *Neurosurgical Review*. 2023;46(1):1-23. [doi:10.1007/s10143-022-01908-y](https://doi.org/10.1007/s10143-022-01908-y)
3. **Huang J**, Galal G, Etemadi M, Vaidyanathan M. Evaluation and Mitigation of Racial Bias in Clinical Machine Learning Models: A Scoping Review. *JMIR Medical Informatics*. 2022;10(5). [doi:10.2196/36388](https://doi.org/10.2196/36388)
4. Huang BB, **Huang J**, Swong KN. Natural Language Processing in Spine Surgery: A Systematic Review of Applications, Bias, and Reporting Transparency. *World Neurosurgery*. 2022;167:156-164. [doi:10.1016/j.wneu.2022.08.109](https://doi.org/10.1016/j.wneu.2022.08.109)
5. Shlobin NA, **Huang J**, Lam SK. Health Literacy in Neurosurgery: A Scoping Review. *World Neurosurgery*. 2022;166:71-87. [doi:10.1016/j.wneu.2022.07.023](https://doi.org/10.1016/j.wneu.2022.07.023)
6. **Huang J**, Shlobin NA, DeCuyper M, Lam SK. Deep Learning for Outcome Prediction in Neurosurgery: A Systematic Review of Design, Reporting, and Reproducibility. *Neurosurgery*. 2022;90(1):16-38. [doi:10.1227/NEU.0000000000001736](https://doi.org/10.1227/NEU.0000000000001736)
7. **Huang J**, Shlobin NA, Lam SK, DeCuyper M. Artificial Intelligence Applications in Pediatric Brain Tumor Imaging: A Systematic Review. *World Neurosurgery*. 2022;157:99-105. [doi:10.1016/j.wneu.2021.10.068](https://doi.org/10.1016/j.wneu.2021.10.068)
8. **Huang J**, Rabin EE, Stricsek G, Swong KN. Outcomes and Complications of Minimally Invasive Transforaminal Lumbar Interbody Fusion in the Elderly: A Systematic Review and Meta-Analysis. *Journal of Neurosurgery: Spine*. 2021;12:1-12. [doi:10.3171/2021.7.SPINE21829](https://doi.org/10.3171/2021.7.SPINE21829)

Case Reports:

1. LoPresti MA, **Huang J**, Shlobin NA, Curry DJ, Weiner HL, Lam SK. Vagus nerve stimulator revision in pediatric epilepsy patients: a technical note and case series. *Child's Nervous System*. 2022. [doi:10.1007/s00381-022-05769-0](https://doi.org/10.1007/s00381-022-05769-0)
2. **Huang J**, Murthy KN, Franz C, Samet J, Deshmukh S, Swong KN. Anterior Interosseous Nerve Neuropathy in a Patient with Spinal Cord Injury: Case Report and Literature Review. *Spinal Cord Series and Cases*. 2022;8(1):61.

Oral Presentations:

1. **Huang J**, Wittbrodt M, Teague C, et al. Efficiency Impacts of Draft Computed Tomography Reporting by Generative AI. *RSNA 2025*. Chicago, IL. Dec 2025. ****Awarded the RSNA 2025 Trainee Research Prize***
2. **Huang J**, Rabin EE, Chandrasekar S, Selner AN, Swong KN. Utility of Intraoperative Neuromonitoring in Minimally Invasive Posterior Lumbar Fusion. *Spine Summit 2022*. Las Vegas, NV. Feb 2022.
3. **Huang J**, Rabin EE, Swong KN. Outcomes and Complications of Minimally Invasive Transforaminal Lumbar Interbody Fusion in the Elderly: A Systematic Review and Meta-Analysis. *SMISS Annual Forum 2021*. Las Vegas, NV. Oct 2021.

Peer-Reviewed Abstracts:

1. Rabin EE, Mi X, **Huang J**, et al. Age-Stratified Survival Analyses of Pharmacological Treatments in Patients with Glioblastoma. *Society for Neuro-Oncology Annual Meeting 2022*. Toronto, Canada. August 2022.
2. Rabin EE, Mi X, **Huang J**, et al. Age-Stratified Survival Analyses of Co-Morbidities in Patients with Glioblastoma. *Society for Neuro-Oncology Annual Meeting 2022*. Toronto, Canada. August 2022.
3. **Huang J**, Konopek N, Moonjely J, Fawzi A. Semi-Automated Blood Velocity Determination Using Adaptive Optics Scanning Laser Ophthalmoscopy XT Images. *ARVO 2022 Annual Meeting*. Denver, CO. May 2022.
4. **Huang J**, Rabin E, Chandrasekar S, Swong K. Minimally Invasive Transforaminal Lumbar Interbody Fusion in Elderly Patients. *American Association of Neurological Surgeons Annual Scientific Meeting*. Philadelphia, PA. April 2022.
5. **Huang J**, Rabin E, Chandrasekar S, Shlobin N, Swong K. Impact of Resident Involvement in Minimally Invasive Transforaminal Lumbar Interbody Fusion. *American Association of Neurological Surgeons Annual Scientific Meeting*. Philadelphia, PA. April 2022.
6. Ellis E, **Huang J**, Drumm M, Rai S, Yerneni K, et al. Risk Factors for Postoperative Seizures and Long-Term AED Use following Meningioma Resection. *American Association of Neurological Surgeons Annual Scientific Meeting*. Philadelphia, PA. April 2022.
7. Gajjar A, **Huang J**, Shlobin N, Le A, Jain A, Dahdaleh N. Top Social Media Influencers in Neurosurgery: An Analysis of Twitter Data. *American Association of Neurological Surgeons Annual Scientific Meeting*. Philadelphia, PA. April 2022.
8. **Huang J**, Rossen J, Rahmani B, Mets-Halgrimson R. Pediatric Eyelid and Canaliculal Lacerations: Epidemiology and Outcomes, *AAPOS 2022 Annual Meeting*. Scottsdale, AZ. Mar 2022.
9. Nadimpalli S, **Huang J**, Ralay-Ranaivo H, Mets-Halgrimson R. Association of Fusional Amplitudes with Surgical Outcomes in Pediatric Strabismus, *AAPOS 2022 Annual Meeting*. Scottsdale, AZ. Mar 2022.
10. Kim E, Stec M, Shaikh N, **Huang J**, Ralay Ranaivo H, Mets-Halgrimson R. Refractive Changes in Children in the Chicagoland Area during the COVID-19 Pandemic. *AAPOS 2022 Annual Meeting*. Scottsdale, AZ. Mar 2022.

11. Domingo JE, Soni P, Galal GO, Mukhin V, **Huang J**, et al. Natural Language Processing for Detection and Reporting of Findings Requiring Follow-Up in Radiology Reports. *Diagnostic Error in Medicine 14th Annual International Conference*. Virtual Conference, Oct 2021.
12. **Huang J**, Shlobin NA, DeCuyper M, Lam SK. Deep Learning for Outcome Prediction in Neurosurgery: A Systematic Review. *American Association of Neurological Surgeons Annual Scientific Meeting*. Orlando, FL. Aug 2021.
13. **Huang J**, LoPresti MA, Shlobin NA, Lam SK. Vagus Nerve Stimulator Revision in Pediatric Epilepsy Patients: A Case Series. *American Association of Neurological Surgeons Annual Scientific Meeting*. Orlando, FL. Aug 2021.

INVITED TALKS

Educational Session, RSNA 2026

December 1, 2026

Invited Speaker

Chicago, IL

- “Seeing the Bigger Picture: Expanding the Radiologist’s Reach with Draft Reporting”

Grand Rounds, Northwestern University Department of Ophthalmology

July 24, 2026

Medical Student Speaker

Chicago, IL

- “The (Visually) Dark Triad of Susac Syndrome”

AIMS Summit 2025

March 13-14, 2025

Invited Speaker & Panelist

Chicago, IL

- “Foundational Generative AI Models in Medical Imaging”

Grand Rounds, Northwestern University Department of Anesthesiology

September 13, 2024

Invited Speaker

Chicago, IL

- “ARIES: Generative AI for Medical Imaging Interpretation”

Dell Technologies World 2024

May 20-23, 2024

Invited Speaker & Panelist

Las Vegas, NV

- “Charting the Generative AI landscape in healthcare” & “How to Choose the Right Server for Current and Future AI Acceleration Needs”